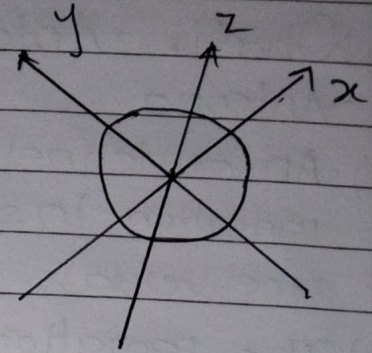
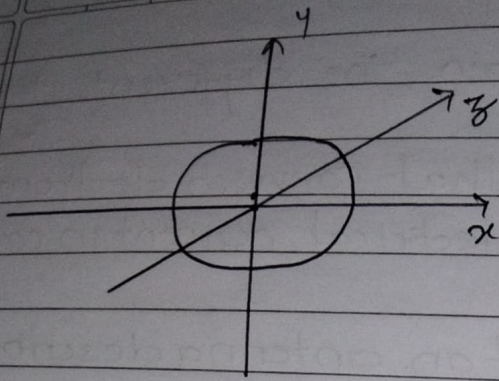


Q.1] What is Antenna & Explain the different types of Antenna.

- 1) An antenna is a device that converts electromagnetic radiation in space into electrical current in conductor or vice-versa.
- 2) The radiation pattern of an antenna describes the relative strength of the radiated field in various directions from the antenna, at a constant distance.
- 3) In reality the radiation pattern is 3 dimensional but usually the measured radiation patterns are a two dimensional slice of the three dimensional pattern.
- 4) There are various types of Antenna discussed below:

A] Isotropic Antenna :-

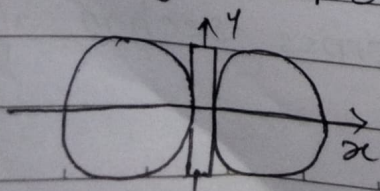
- i) An Isotropic Antenna is a theoretical antenna that radiates its power uniformly in all directions.
- ii) In other words, theoretical isotropic antenna has a perfect 360 degree spherical radiation pattern.
- iii) It is an ideal antenna which radiates equally in all directions & has a gain of 1 (0 dB) i.e. zero gain and zero loss.
- iv) It is used to compare the power level of given antenna to the theoretical isotropic antenna.
- v) Antenna can be broadly classified as omnidirectional and directional antenna.
- vi) Fig. shows a 2 dimensional cross-section of the real 3 dimensional pattern.



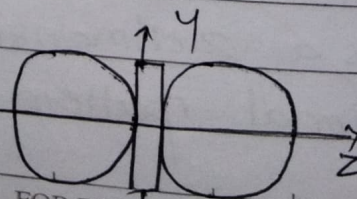
Radiational pattern of isotropic antenna

By Omnidirectional Antenna :-

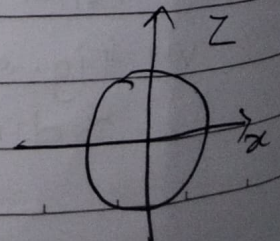
- i) Unlike Isotropic Antenna, Dipole Antennas are real antennas.
- ii) The dipole radiation pattern is 360 degrees in the horizontal plane and approximately 75 degrees in the vertical plane.
- iii) It is also called the non-directional antenna because it does not favor any particular direction.
- iv) Dipole Antennas are said to have a gain of 2.14 dB which is in comparison to an isotropic antenna.
- v) The higher the gain of the antenna, the smaller the vertical beam width is.
- vi) This type of antenna is useful for broadcasting signals to all points of the compass or when signals from all points.



Side view (xy plane)



FOR EDUCATIONAL USE
Side view (zy plane)



Top view (xz plane)

Q.27 → Compare all Mobile Generation i.e. 1G, 2G, 3G, 4G and 5G.

Parameter	1G	2G	3G	4G	5G
1) Introduced	1980	1993	2001	2009	currently under roll out 2020.
2) Technology	Amps	D-Amps IS-95, GSM	W-CDMA CDMA2000 UWC-136 TD-CDMA DECT	LTE, Wimax	LTE Advanced, OMA & No MA, WNW -W.
3) Multiplexing	FDMA	TDMA/CDMA	CDMA	CDMA	CDMA
4) Switching Type	Circuit Switching	Circuit Switching for voice	Packet Switching	All packet Switching	All packet Switching
5) Speed	2.4 kbps to 14.4 kbps	14.4 kbps	3.1 mbps	100 mbps	> 10 Gbps
6) Band width	Analog	25 MHz	25 MHz	100 MHz	60 GHz
7) Hand over	NA	Horizontal	Horizontal	Horizontal/ Vertical	Horizontal/ Vertical

Q.37 Describe GSM in detail.

- 1) Globem System for Mobile Communication (GSM) is the most successful digital Mobile telecommunication system in the world today.
- 2) It is used by over 1000 million people in more than 190 countries.
- 3) The primary goal of GSM was to provide a mobile phone system that allows users to roam throughout Europe and provide voice service.
- 4) Globem System for Mobile Communication (GSM) is a globally accepted standard for digital cellular Communication. It is an ETSI standard for 2G pan European digital cellular system with international roaming.
- 5) The basic version of GSM (ie. GSM900) was founded in 1982.
- 6) Now it is the most successful Mobile Communication System in the world and over 1.2 billion users use the system.
- 7) GSM is the 2nd generation 2G system replacing the 1st generation 2G S-Analog system.
- 8) The initial version of GSM was designed in Europe using 890-915 MHz for Uplink and 935-960 MHz for downlink. This system is called GSM900.

- 9) Another version of GSM called digital cellular system 1800 (DCS 1800) were, 1710-1785 MHz for Uplink and 1805-1880 MHz for downlink.
- 10) GSM at 1900 MHz used in us is called PCS 1900.

Modification & Derivatives of GSM:

The system Evolution of GSM can be divided into 3 points.

- Phase 1 (1981-1994): The basic version of the GSM system was in operation.
- Phase 2 (1984-1995): The system specification was verified in order to allow future gradual modification & new improvement.
- Phase 3 (from 1995): The modification to the original GSM 900 are being introduced.

Q.47) What are the characteristics of SIM.

→ - If you mean SIM (Subscriber Identity Module) in Mobile telecom, here are its key characteristics in simple terms:

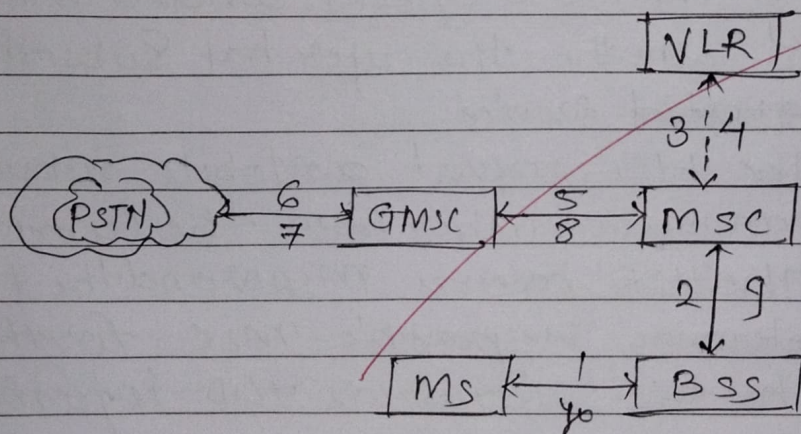
- 1) Unique identification :- you can each sim have a unique IMSI (International Mobile Subscriber Identity)
- 2) Authentication & Security :- Stores secret keys used to authenticate the user with the Mobile networks. Enabled encrypted communication helping prevent fraud and eavesdropping.
- 3) Subscriber Information Storage :-
 - Holds data like:
 - Phone number
 - Network authorization detail
 - Sometimes contacts and SMS.
- 4) Small Standardization form :- Come in different size standard gsm sim, micro-sim, Nano sim and now eSim.
- 5) Operator specific :- Issued and managed by Mobile network operator, not the phone manufacturer.

Q.5] Explain Mobile Oriented and terminated call.

→ It is much simpler to perform a mobile oriented call (MOC) compared to MTC. This follows the following steps:

Step 1: The MS transmits the request for a new connection. This is realized by the MS sending a random access burst on RACH logical channel.

Step 2: The BSS forwards this request to the MSC.
Step 3, 4, 8 → MSC then checks if this user is allowed to setup a call with the requested service.



Mobile Oriented call

Step 3, 8 :- The MSC checks the availability of resources through Gsm Network and into the PSTN

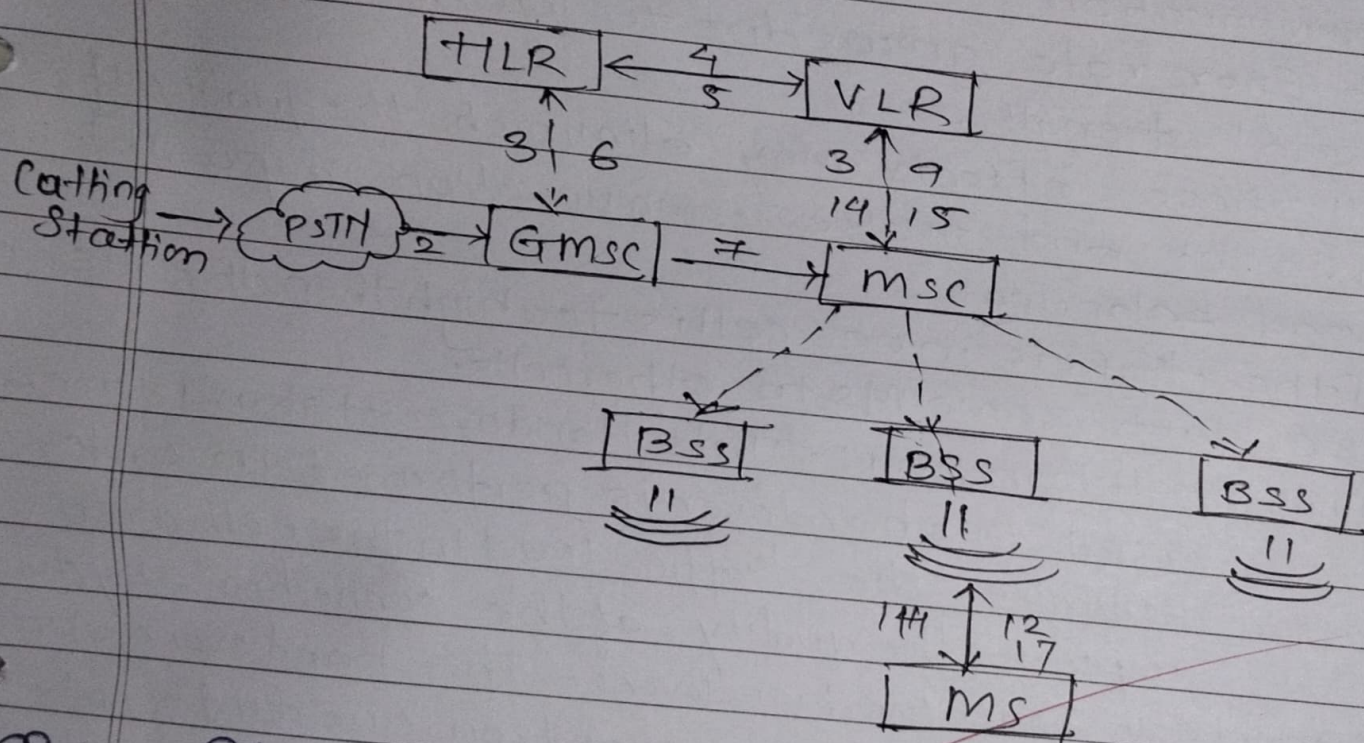
Step 9, 10 → If all resources are available then MSC setup connection between MS and the fixed network.

Mobile Terminated Call (MTC):-

This is the situation where a terminal from a fixed network call a mobile station. This involves the following steps:

- step 1:- A PSTN user client dials the Ph.No. of a Gsm subscriber. The fixed network (PSTN) notices that the no. is of the Gsm network & forward call setup to the Gateway MSC.
- step 2: GmSC identifies the HLR for the subscriber & signals the call setup to the HLR.
- step 3: The HLR now checks whether the no. exists and whether the user has subscribed to the requested service.
- step 4:- The HLR request a Mobile Subscriber roaming number from the current VLR.
- step 5:- The HLR receives MURN and the HLR can determine responsible MSC for the ms.
- step 6: The HLR forwards this information to GmSC.
- step 7:- The GmSC forward this information to GmSC.
- step 8: The GmSC forward call setup request to the MSC.
- step 9: The MSC first request the current status of the MS from the VLR.
- step 10: If the MS is available the MSC initiates paging in all cells.

Step 11:- The BU of all BSS / transmits this par
 signal to the MS.
 step 12, 13:- The ms answers
 step 14, 15:- The VLR does security checks
 step 16, 17:- The Connection is Setup.



Q.6) What are the different types of Handover in

- 1) When a mobile user is engaged conversation th
 is connected to the BTS via radiolink.
- 2) If the mobile users move to the coverage area
 another BTS, the radio link to the old BTS
 eventually disconnected and radio link to the new
 established to the conversation.
- 3) The number of handover to be performed dep
 on two factor:

1] Cell size

2] Speed of MS

- There are two Basic reasons for handover!

I] MS moves out of the range of BTS:-
- As a Mobile station is moved out of the range of BTS, the received signal level requirement of communication.

- The Error rate grows due to interference & low signal strength.

- All these effects may diminish the quality of radiolink and make communication impossible.

2] Load balancing!

- If the traffic in one cell is too high then the BSC shifts some MS to other cells.

I] Intra cell handover:- This handover takes place within a cell. This handover is performed in order to optimize the traffic load in the cell and improve the quality of the connection.

II] Inter-cell, Inter BSC handover:- This handover takes place when a mobile station moves from one cell to another cell but stays with the control of the same BSC.

III] Inter-BSC, intra MSC, handover:- This handover takes place between two cells managed by different BSCs. This handover is controlled by the MSC.

IV] Inter MSC handover:- Inter MSC handover takes place between two cells belonging to different MSCs. Both MSCs perform the handover together.

Handover